VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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Major Project Phase-II Report on

"Smart AgroSpray: AI-Driven Precision Pesticide Spraying with IoT"

Submitted in partial fulfillment for the Major Project Phase-II (BCS786) course of Seventh Semester of Bachelor of Engineering in Computer Science & Engineering during the academic year 2025-26.

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~~ CERTIFICATE ~~

*Certified that the project work entitled "**Smart AgroSpray: AI-Driven Precision Pesticide Spraying with IoT**" is a bonafide work carried out by **Rangaswamy S G** (4MH22CS129), **Raviteja K S** (4MH22CS131), **Durgesh Gowda H B** (4MH23CS403) & **Adithya M B** (4MH23CS400) for the Major Project Phase-II (BCS786) of Seventh Semester in Computer Science & Engineering under Visvesvaraya Technological University, Belagavi during academic year 2025-26.*

It is certified that all corrections/suggestions indicated for Internal Assignment have been incorporated in the report. The report has been approved as it satisfies the academic requirements in respect of the final year major project work.

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~~~ ABSTRACT ~~~

In this project, an intelligent AI and IoT-integrated robotic system titled “Smart AgroSpray: AI-Driven Precision Pesticide Spraying with IoT” has been developed to overcome the challenges of excessive pesticide usage and delayed disease detection in tomato cultivation. The system employs a YOLOv11 deep learning model trained on healthy and diseased tomato leaf images to identify infections in real time through a camera connected to a laptop. When a diseased leaf is detected, a command signal is transmitted via Bluetooth (HC-05) to an ESP8266 microcontroller mounted on a mobile robot, which then activates a relay-controlled submersible pump to spray pesticide for a predefined duration before automatically switching off. If no disease is detected, spraying does not occur. The robot’s movement is controlled through the Blynk mobile application, enabling navigation in multiple directions, while an IR sensor ensures obstacle detection for safe operation. This seamless AI–IoT integration ensures that pesticide is applied only when necessary, minimizing chemical wastage, reducing human exposure, and promoting environmentally sustainable farming. Experimental results demonstrate high detection accuracy of about 90–92% and reliable real-time performance. Overall, the system provides a cost-effective, scalable, and efficient smart agriculture solution that combines artificial intelligence, computer vision, and IoT-based robotic control to enhance precision pesticide application and improve crop productivity.

~~~~~ Program Outcomes Mapping ~~~~~

Sl. No	Program Outcomes / Program Specific Outcomes	Mapping (Scale of 1 to 3)
PO-01	**Engineering Knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.	**3**
PO-02	**Problem Analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.	**3**
PO-03	**Design/Development of Solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.	**3**
PO-04	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.	**2**
PO-05	**Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.	**3**
PO-06	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.	**2**
PO-07	**Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.	**3**
PO-08	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.	**1**
PO-09	**Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.	**3**
PO-10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	**2**
PO-11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	**2**
PO-12	**Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.	**2**
PSO-1	**Apply** software engineering practices and strategies in diversified areas of computer science for solving problems using open-source environments.	**3**
PSO-2	**Develop** suitable algorithms and codes for applications in areas of cognitive technology, computer networks with software engineering principles and practices.	**3**

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CHAPTER - 1

INTRODUCTION

1.1 Overview

Agriculture is one of the most crucial sectors sustaining the global economy, particularly in India, where a major portion of the population depends on farming for livelihood. Among all cultivated crops, tomato is a highly consumed and economically valuable crop due to its nutritional value and year-round demand. However, tomato plants are extremely susceptible to several leaf diseases such as early blight, late blight, leaf spot, and mosaic diseases, which can drastically affect yield and fruit quality if not detected and treated promptly.

Conventional methods for detecting crop diseases rely on manual inspection by farmers or agricultural experts. This process is time-consuming, labour-intensive, prone to human error, and infeasible for large-scale monitoring. Similarly, traditional pesticide spraying involves applying chemicals uniformly across entire fields without identifying infected areas, leading to unnecessary chemical wastage, increased production costs, and potential environmental harm.

With recent advancements in artificial intelligence (AI), computer vision, and the Internet of Things (IoT), precision agriculture has emerged as an innovative approach to address these challenges. Deep learning models such as YOLO (You Only Look Once) enable real-time object detection and classification, allowing for automated disease identification from live plant images. Integrating such AI systems with IoT-based robotic platforms enables immediate, targeted spraying action — achieving real-time, automated precision in pest control.

This project, titled “Smart AgroSpray: AI-Driven Precision Pesticide Spraying with IoT,” aims to implement such an integrated system for tomato leaf disease detection and automated pesticide application, reducing manual effort, pesticide wastage, and ecological damage while increasing efficiency and sustainability in farming.

1.2 Motivation

Agriculture in India continues to face challenges related to disease outbreaks, inefficient pesticide usage, and limited access to modern technology. Farmers often rely on traditional visual inspection and manual pesticide application, which are neither time-efficient nor environmentally safe. Continuous exposure to pesticides also poses serious health risks to farmers and causes contamination of soil and water sources.

The motivation behind this project arises from the need to bridge the gap between emerging AI technologies and their real-world agricultural applications. By using computer vision for early disease detection and integrating it with an automated IoT-based robotic sprayer, farmers can treat diseased plants accurately and efficiently. This reduces human effort, pesticide consumption, and environmental impact.

The project is also driven by the goal of creating a cost-effective precision agriculture solution that can be adopted by small and medium-scale farmers. Using affordable components such as the ESP8266 microcontroller, HC-05 Bluetooth, and Blynk application, the system provides an accessible approach to smart farming without requiring complex infrastructure or technical expertise.

1.3 Aim and Objectives

Aim

To develop an AI and IoT-integrated robotic system capable of real-time tomato leaf disease detection and automated pesticide spraying for precision agriculture.

Objectives

1. To train and deploy a YOLOv11 deep learning model for accurate real-time classification of healthy and diseased tomato leaves.
2. To design a Blynk-controlled four-wheel robotic platform using the ESP8266 microcontroller for field mobility.
3. To establish Bluetooth-based communication between the vision system and the robot to trigger automated pesticide spraying upon disease detection.
4. To minimize pesticide wastage and human exposure through selective, automated spraying.
5. To promote sustainable farming by integrating artificial intelligence, computer vision, and IoT-based control.

1.4 Scope of the Project

The scope of Smart AgroSpray extends across multiple dimensions of precision agriculture. The current prototype demonstrates real-time detection and automated spraying for tomato crops but can be scaled to handle multiple crop types and diseases by retraining the AI model with larger datasets.

The system's modular design allows easy upgrades to cloud-based communication, GPS-enabled navigation, and advanced obstacle avoidance mechanisms. Furthermore, it can be extended to integrate other agricultural operations such as fertilizer application, irrigation control, and environmental monitoring.

By combining AI, IoT, and robotics, the system contributes to the development of smart farming ecosystems, offering improved efficiency, reduced pesticide usage, and enhanced crop quality while supporting environmental sustainability and farmer safety.

1.5 Organization of the Report

The report is organized into the following chapters:

1. Chapter 1 – Introduction: Provides an overview, motivation, objectives, and scope of the project.
2. Chapter 2 – Literature Survey: Discusses existing research, technologies, and systems related to AI-based plant disease detection and IoT-enabled agriculture.
3. Chapter 3 – System Requirement Analysis: Explains system feasibility, functional and non-functional requirements, and hardware/software specifications.
4. Chapter 4 – System Design: Presents system architecture, data flow, and algorithmic design of the proposed solution.
5. Chapter 5 – Implementation: Details the modules, integration process, and step-by-step implementation of both hardware and software components.
6. Chapter 6 – Results and Analysis: Describes experimental results, accuracy analysis, and system performance evaluation.
7. Chapter 7 – Testing and Validation: Discusses test cases, testing methodology, and validation results of the system.

CHAPTER – 2

LITERATURE SURVEY

2.1 Overview of Existing Systems

Traditional agricultural disease management systems primarily rely on manual visual inspection by farmers or experts to identify crop infections. This conventional method is not only time-consuming and subjective but also prone to human error, especially in large-scale farms where continuous monitoring is impractical. Additionally, the practice of uniform pesticide spraying across the entire field results in excessive chemical usage, leading to soil degradation, environmental pollution, and potential health hazards for farmers.

To address these limitations, early research in digital agriculture introduced image processing-based detection systems using handcrafted features such as color, texture, and shape. Although these methods provided basic detection capabilities, their accuracy was limited under varying lighting and environmental conditions.

The evolution of machine learning and deep learning has transformed agricultural disease detection by enabling automatic feature extraction and real-time classification. Models like CNNs (Convolutional Neural Networks) and YOLO (You Only Look Once) have been successfully used to detect plant diseases from leaf images with high accuracy. Concurrently, IoT-based systems have been employed for automated control of irrigation, temperature, humidity, and pesticide spraying.

However, most existing systems focus on either AI-based disease detection or IoT-based actuation, treating them as separate modules. Few solutions effectively integrate AI-driven decision-making with real-time IoT action, limiting their practical deployment in field conditions. The need for an integrated AI–IoT framework for precision agriculture remains a critical research gap that this project aims to address.

2.2 Summary of Related Research Papers

1. Costa et al. (2024) – "An AI-Driven Architecture for Precision Agriculture: IoT, Machine Learning, and Digital Twin Integration for Sustainable Crop Protection" (IEEE Big Data, 2024) This paper presents an AI-based architecture combining IoT sensors, machine learning, and digital twin technology for adaptive crop protection. A "Digital Tree" model simulates olive trees to predict spray behavior and environmental interactions. Real-time IoT data refines spraying parameters dynamically, minimizing off-target pesticide losses and improving sustainability. The

study emphasizes that AI–IoT integration enables intelligent, data-driven decision-making for precision farming.

2. Narendra Babu et al. (2023) – "Internet of Things (IoT) Based Pesticide Spraying Robot – A Revolution in Smart Farming" (Indian Journal of Science and Technology, 2023)
   The authors propose an IoT-controlled ARM7-based pesticide spraying robot aimed at reducing pesticide wastage and human exposure. Using Bluetooth control and HC-05 communication, the system activates a DC pump and motor driver to spray pesticide precisely based on farmer input. The robot demonstrates low power consumption (3.3 V), extended spraying range (50 m), and cost-effectiveness (< USD 250), highlighting the feasibility of low-cost smart farming robots for Indian farmers.
3. Thakur et al. (2022) – "Pesticide Sprayer for Agricultural Purpose Based on IoT Technology" (IEEE ICOSEC, 2022)
   This study integrates AI-based plant health detection (SVM, KNN, CNN) with an IoT-enabled robotic sprayer controlled via Blynk and GSM SIM808. The robot captures live images using a webcam, classifies diseased tomato leaves using CNN (~90% accuracy), and initiates spraying only when infection is detected. A robotic arm adjusts nozzle distance based on ultrasonic sensor data, optimizing pesticide usage. The prototype demonstrates efficient AI–IoT collaboration for real-time disease detection and selective pesticide application.
4. Krishnaleela et al. (2024) – "Multi-Purpose Agricultural Pesticide Spraying Robot Using IoT" (IEEE ICAITPR, 2024)
   This paper introduces a Bluetooth-controlled multipurpose agricultural robot capable of seed planting, fertilizer, and pesticide spraying. The machine employs Arduino-based control, dual relay modules, and gear motors to execute various farming operations. It emphasizes resource optimization, reduced labor dependency, and multipurpose field use. The research highlights how IoT-enabled automation can unify multiple farm functions, making agriculture more sustainable and efficient.
5. Gandham Sandhya et al. (2023) – "Integrating Technology for Sustainable Agriculture: Enhancing Crop Productivity while Minimising Pesticide Usage using Image Processing & IoT" (IEEE ICESC, 2023)
   This paper presents an integrated system combining image processing, IoT sensors, and drone technology to improve crop productivity while reducing pesticide use. Drones capture field images, which are processed using k-means clustering and GLCM features, and classified by an SVM model to detect diseased regions. Upon detection, drones perform targeted pesticide spraying, guided by IoT-based environmental data such as soil moisture and temperature. The system achieved 95–97% accuracy, minimizing chemical wastage and promoting data-driven

sustainable farming through precise disease detection and controlled pesticide application.

2.3 Conclusion of Literature Review

From the reviewed research, it is evident that the combination of artificial intelligence and IoT offers immense potential for precision agriculture. While individual advancements in AI-based disease detection and IoT automation have shown success, most previous works treated these technologies as independent systems.

The review emphasizes that integrating AI's decision-making capabilities with IoT's actuation control enables a closed-loop, real-time precision farming mechanism. Such integration minimizes human intervention, reduces pesticide wastage, and allows data-driven disease management. The insights from prior research form the theoretical foundation for developing Smart AgroSpray, an end-to-end AI–IoT system for intelligent, automated pesticide spraying.

2.4 Existing System and Limitations

The existing agricultural systems primarily employ manual disease inspection and uniform pesticide application, which pose several challenges:

1. Lack of precision: Spraying chemicals over entire fields without identifying diseased areas leads to wastage and environmental pollution.
2. Human dependency: Continuous monitoring requires skilled labor and is prone to fatigue and human error.
3. Health hazards: Farmers are exposed to toxic pesticides during manual spraying.
4. High operational cost: Unnecessary use of chemicals increases input costs for farmers.
5. Low scalability: Manual inspection cannot be applied effectively on large-scale farms.

Although modern computer vision-based systems can detect diseases automatically, they often do not perform real-time actions after detection. Similarly, IoT-enabled sprayers can automate spraying but lack decision-making capability to determine where and when to spray.

Thus, the key limitation of existing systems is the absence of AI-driven real-time actuation. The Smart AgroSpray project overcomes this limitation by integrating a YOLOv11-based disease detection model with an IoT-enabled robotic sprayer, forming a unified intelligent system that detects, decides, and acts autonomously — a significant advancement in the domain of AI-driven precision agriculture.

CHAPTER – 3

SYSTEM REQUIRMENT ANALYSIS

3.1 Overview of Proposed System

The proposed project, Smart AgroSpray: AI-Driven Precision Pesticide Spraying with IoT, aims to create a cost-effective and intelligent system for precision pesticide application in tomato cultivation. The system integrates Artificial Intelligence (AI) for disease detection and IoT-based robotic actuation for automated spraying, ensuring that pesticide is applied only to diseased areas.

The overall system functions as follows: live images of tomato plants are captured using a camera connected to a laptop. The YOLOv11 deep learning model, trained on a dataset of healthy and diseased tomato leaf images, analyzes each frame in real time. If the model detects a diseased leaf, it transmits a command via Bluetooth (HC-05) to the ESP8266 microcontroller mounted on a mobile robot. The microcontroller then activates a relay-controlled submersible pump to spray pesticide for a specific duration and automatically turn it off.

The robot is navigated manually using the Blynk mobile application, which sends directional control signals (forward, backward, left, right) through Wi-Fi. Additionally, IR sensors detect obstacles to prevent collisions. This integration of AI decision-making with IoT hardware enables a closed-loop precision spraying system, reducing manual intervention, pesticide wastage, and environmental hazards.

The overall working of the proposed system is shown in Figure 3.1

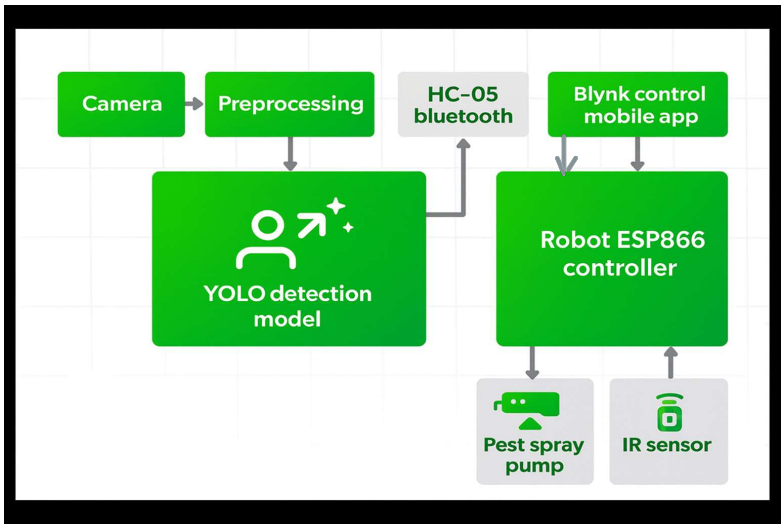

Figure 3.1: Proposed System Architecture of Smart AgroSpray

Figure 3.1 shows how the AI and IoT modules interact to enable precision pesticide spraying. The camera captures real-time images of tomato plants, which are preprocessed and analyzed using the YOLO detection model to identify diseased leaves. Once a disease is detected, the model transmits a command via the HC-05 Bluetooth module to the ESP8266 microcontroller, which triggers the pesticide spray pump through a relay connection. The Blynk mobile application enables manual navigation of the robot, while the IR sensor assists in obstacle detection, ensuring safe and efficient operation in field conditions.

3.2 Feasibility Study

The feasibility study determines the practicality, sustainability, and effectiveness of implementing the Smart AgroSpray system in real-world agricultural environments. It evaluates whether the proposed design can be successfully developed, deployed, and maintained with available resources while achieving its intended objectives. The analysis considers four aspects — technical, economic, operational, and environmental feasibility.

3.2.1 Technical Feasibility

The project is technically feasible because it relies on readily available, low-cost, and well-documented hardware components, ensuring smooth development and integration. The ESP8266 microcontroller supports both Wi-Fi and serial communication, enabling seamless data exchange between the AI module and the spraying robot. The YOLOv11 deep learning model is lightweight and optimized for real-time inference, allowing it to run effectively on mid-range computing systems without requiring GPUs.

Integration with the Blynk IoT platform provides remote control, status monitoring, and live feedback through a user-friendly mobile interface. The HC-05 Bluetooth module ensures reliable short-range communication between the AI system and microcontroller, while open-source software frameworks like Arduino IDE, Python, and OpenCV simplify implementation and debugging.

Additionally, the modular architecture allows future upgrades such as cloud connectivity, sensor-based decision-making, or autonomous navigation. Hence, the system is both technically achievable and scalable for extended agricultural applications.

3.2.2 Economic Feasibility

The proposed Smart AgroSpray system is economically viable, as it utilizes cost-efficient electronic components that are easily available in the local market. The total hardware cost of the prototype is

considerably lower than that of commercial agricultural robots or automated spraying systems, making it accessible to small and medium-scale farmers.

The system also provides long-term cost savings by reducing unnecessary pesticide consumption through selective spraying. By minimizing chemical use and manual labor, the project decreases recurring operational expenses while improving crop yield and quality.

In addition, the maintenance cost of components such as the ESP8266, relay modules, and submersible pump is minimal, and all parts can be easily replaced without specialized tools. The economic benefits thus outweigh the initial investment, ensuring high return on investment (ROI) and long-term financial feasibility for field deployment.

3.2.3 Operational Feasibility

Operational feasibility ensures that the proposed system can be efficiently used by end-users (farmers) without requiring extensive technical training. The Blynk mobile application interface allows intuitive manual control of robot movement and system monitoring in real time. The spraying operation is fully automated once the AI model detects diseased leaves, eliminating the need for manual pesticide handling.

The system's components are compact and portable, allowing easy transportation across fields. It operates effectively in standard agricultural environments, and maintenance procedures are simple enough for non-technical users. The AI–IoT coordination ensures continuous and synchronized functioning between disease detection, command transmission, and spraying response.

The system also incorporates fail-safe mechanisms, such as obstacle detection via IR sensors, ensuring operational safety. Therefore, it is both user-friendly and reliable, capable of continuous operation even under semi-supervised conditions.

3.2.4 Environmental Feasibility

The Smart AgroSpray system promotes sustainable and eco-friendly agricultural practices by ensuring that pesticide spraying occurs only on diseased plants, significantly reducing overall chemical usage. This selective spraying approach prevents the accumulation of harmful residues in soil and water, thereby minimizing environmental pollution.

By reducing unnecessary chemical exposure, the system helps maintain ecological balance, protecting beneficial insects, soil microorganisms, and surrounding biodiversity. Moreover, automated

spraying reduces direct human contact with toxic substances, lowering the risk of skin irritation, respiratory illness, or long-term health effects for farmers.

The use of energy-efficient components such as low-voltage motors and pumps also contributes to sustainability by conserving power. Overall, the project aligns with the principles of sustainable development and smart agriculture, offering a practical solution that enhances productivity while safeguarding the environment.

3.3 Functional Requirements

Functional requirements define the specific operations and behaviors the system must perform to achieve its intended purpose. The proposed Smart AgroSpray system is designed to detect diseased tomato leaves using AI and perform targeted pesticide spraying through IoT-based actuation. The key functional requirements are as follows –

1. Image Acquisition Module – Captures live images or video frames of tomato plants using a camera connected to the laptop or processing unit. This serves as the primary input for disease detection.
2. AI Detection Module – Processes each image frame using the YOLOv11 deep learning model to identify diseased leaves in real time with high accuracy.
3. Command Transmission – Sends detection signals from the AI model to the IoT hardware through the Bluetooth HC-05 module to initiate spraying.
4. IoT Control Module – The ESP8266 microcontroller receives the detection signal and controls the relay circuit, which in turn activates or deactivates the submersible pump.
5. Spraying Mechanism – Operates the submersible pump to spray pesticide selectively on diseased plants for a predefined duration and automatically stops afterward.
6. Mobile Control (Blynk) – Provides manual control options through the Blynk mobile application for robot navigation in multiple directions and allows real-time monitoring of spraying status.
7. Obstacle Detection – Utilizes IR sensors to detect obstacles in the robot's path, preventing collisions and ensuring smooth operation in the field.
8. Power Management – Ensures stable and regulated power supply to all system components, maintaining safe voltage levels for continuous operation.

3.4 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance constraints, and overall characteristics that determine how efficiently and reliably the system operates. The Smart AgroSpray system must satisfy the following non-functional requirements –

1. Performance – The system should perform real-time detection and spraying operations with minimal latency. The time delay between disease detection and actuation should not exceed one second.
2. Accuracy – The YOLOv11 AI model should maintain a detection accuracy of at least 90% to ensure precise identification of diseased leaves and minimize false spraying.
3. Reliability – All hardware and software components should function consistently under normal outdoor conditions such as varying light, temperature, and dust exposure.
4. Scalability – The system design should allow easy expansion to include additional crops, diseases, or sensors without major modifications to the core architecture.
5. Usability – The mobile control interface through the Blynk application should be intuitive, allowing farmers to operate the robot easily with minimal technical knowledge.
6. Security – Communication between the AI module, Bluetooth, and ESP8266 controller should be secured to prevent unauthorized access or interference with device operation.
7. Maintainability – The modular design of both hardware and software should allow easy debugging, component replacement, and system updates when necessary.
8. Portability – The robot should be lightweight, compact, and easily transportable within agricultural fields, allowing smooth operation over uneven terrain.
9. Power Efficiency – The system should utilize power efficiently, ensuring long operational duration with low energy consumption and minimal battery drain.
10. Safety – The spraying mechanism should function only when required to prevent accidental pesticide discharge, ensuring safety for users and the surrounding environment.

3.5 Software and Hardware Requirements

The Smart AgroSpray system integrates both hardware and software components to achieve seamless communication between the AI-based disease detection model and the IoT-controlled pesticide spraying mechanism. The selection of components is based on performance, cost-effectiveness, and compatibility for real-time agricultural operation.

3.5.1 Hardware Requirements

1. ESP8266 Microcontroller – Serves as the central processing unit for the IoT module. It receives the detection command from the AI system via Bluetooth and controls the spraying mechanism through relay activation. It also supports Wi-Fi connectivity, allowing communication with the Blynk mobile platform for robot navigation.

2. HC-05 Bluetooth Module – Establishes serial communication between the laptop running the AI detection model and the ESP8266 microcontroller. It transmits the disease detection signal instantly, enabling real-time synchronization between the AI and IoT units.
3. Submersible Pump – Used as the spraying unit to discharge pesticide onto diseased plants. It operates under low voltage and is activated automatically through the relay when a diseased leaf is detected, ensuring selective pesticide application.
4. Relay Module – Acts as an electrically operated switch that controls the ON/OFF operation of the submersible pump. It receives trigger signals from the microcontroller and isolates the high-voltage pump circuit from the low-voltage control logic for safety.
5. Motor Driver (L298N) – Drives the DC motors that control the robot's movement. It provides bidirectional control, allowing forward, backward, left, and right motion based on signals from the Blynk application.
6. DC Motors and Wheels – Provide mobility to the robotic platform, enabling it to navigate across the field. The motors are compact and energy-efficient, suitable for uneven surfaces in agricultural areas.
7. IR Obstacle Detection Sensor – Detects obstacles in front of the robot and prevents collisions by sending signals to the controller to stop or redirect the robot, ensuring smooth and safe operation.
8. Camera (Webcam or USB Camera) – Captures real-time images of tomato plants for disease detection. The captured frames are analysed by the YOLOv11 model running on the laptop.
9. Chassis Frame – Acts as the structural base of the robot, supporting all components such as motors, sensors, pump, and controller. It ensures stability and compact arrangement of modules for field use.
10. Power Supply (12V and 5V) – Provides regulated power to all components, including motors, sensors, and microcontroller units. A stable power source ensures uninterrupted operation during spraying and navigation.

3.5.2 Software Requirements

1. Python 3.10 or above – Serves as the primary programming language for developing and running the AI detection model. It is used to implement the YOLOv11 algorithm, perform image preprocessing, and send actuation commands.
2. YOLOv11 Framework – Provides the deep learning architecture for detecting diseased tomato leaves. It performs real-time object detection with high accuracy and minimal computational delay.
3. OpenCV Library – Used for image acquisition, processing, and visualization. It helps capture frames from the camera, resize images, and annotate detection results for display.

4. Arduino IDE – Used to program and upload the control logic to the ESP8266 microcontroller. It defines how the relay, motors, and sensors respond to incoming Bluetooth commands.
5. Blynk IoT Platform – Offers a mobile-based interface for manually controlling robot movement. It allows real-time monitoring of robot activity and provides cloud-based connectivity between the user and IoT hardware.
6. Anaconda Navigator and Jupyter Notebook – Provide the environment for dataset handling, model training, and testing. They facilitate efficient development and analysis of the YOLOv11 model.
7. NumPy and TensorFlow Libraries – Support the mathematical and machine learning operations required for model training and inference. They enable GPU optimization and faster computation during detection.
8. Windows 10 or 11 Operating System – Acts as the base environment for development, integration, and execution of all software tools. It supports compatibility with Python, Arduino IDE, and Blynk setup.
9. Firmware for ESP8266 – Embedded software responsible for interpreting Bluetooth and Wi-Fi commands, managing relay control, and ensuring smooth synchronization with the spraying mechanism.

CHAPTER – 4

SYSTEM DESIGN

4.1 Architectural Design

System design plays a crucial role in transforming theoretical concepts into a functional prototype. The Smart AgroSpray architecture defines how the AI-based disease detection module interacts with the IoT-controlled spraying unit, enabling real-time, automated, and selective pesticide application.

The proposed system architecture is divided into three primary layers –

1. AI Layer (Decision-Making Unit) –This layer performs image acquisition, preprocessing, and real-time disease detection using the YOLOv11 deep learning model. The camera continuously captures images of tomato plants and sends them to the laptop for processing. The YOLO model analyzes each frame and classifies it as "healthy" or "diseased." If a diseased leaf is detected, the AI module immediately generates a command signal to activate the spraying mechanism through Bluetooth communication.
2. IoT Control Layer (Execution Unit) –The IoT control layer receives the command signal via the HC-05 Bluetooth module, which is connected to the ESP8266 microcontroller. The ESP8266 interprets the command and triggers a relay module that controls the submersible pump. The relay acts as a digital switch, ensuring electrical isolation between control and power circuits. The pump then sprays pesticide for a fixed duration and automatically turns off. This layer essentially converts the AI detection output into a physical actuation response.
3. Execution and Monitoring Layer (User Interface Unit) –The final layer provides manual control and monitoring features. The Blynk IoT platform allows farmers to control robot movement (forward, backward, left, right) through a smartphone. The IR sensor continuously monitors for obstacles, ensuring that the robot stops or changes direction to avoid collisions. This layer also provides system feedback to the user, displaying real-time operational status such as spraying activity or connection state.

The Smart AgroSpray system follows a layered architecture integrating AI-based decision-making with IoT-based actuation. The AI layer captures real-time images of tomato leaves and analyzes them using the YOLOv11 model to detect disease conditions. Upon detecting a diseased leaf, a control command is generated and transmitted via Bluetooth to the ESP8266 microcontroller. The IoT control layer activates the relay and pesticide pump for selective spraying, while the monitoring layer enables manual robot control through the Blynk application and ensures safe navigation using IR sensors. This

integrated design ensures real-time, precise, and automated pesticide spraying.

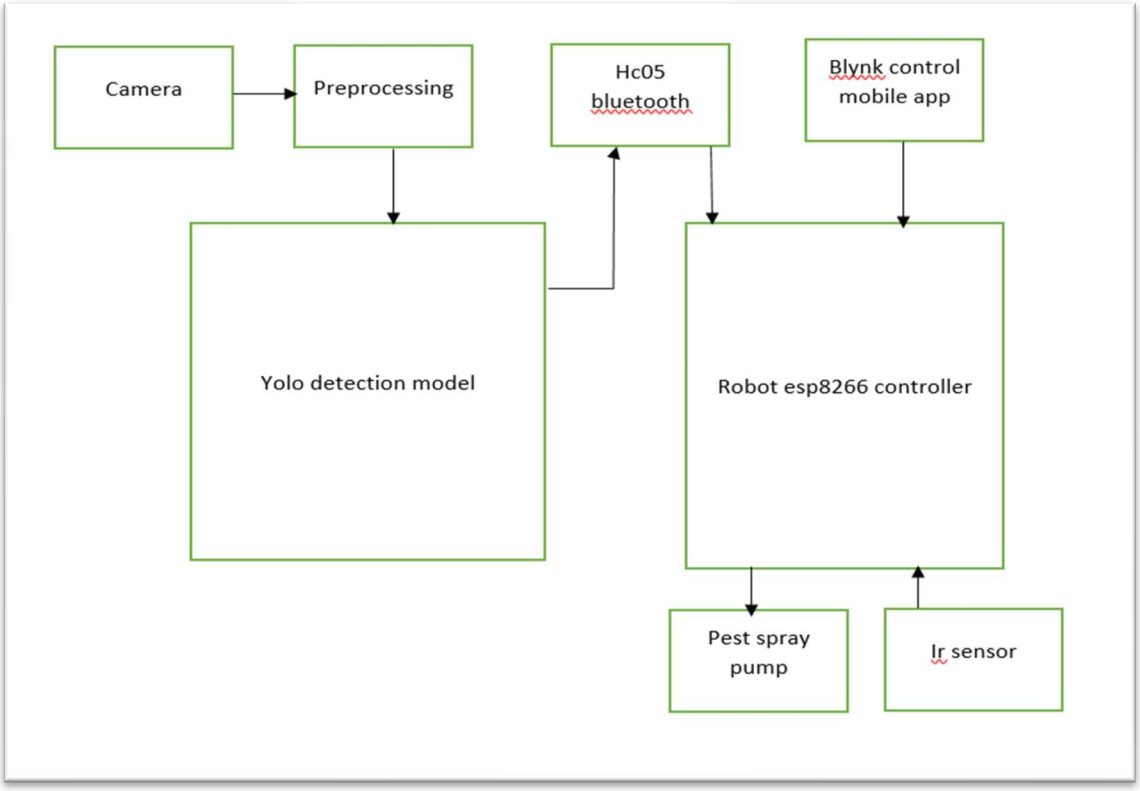

Figure 4.1 – System Architecture of Smart AgroSpray

Figure 4.1 shows the architecture diagram clearly represents the logical and physical communication flow between the system components. The camera captures real-time video, which is processed by the YOLO model to detect diseased leaves. Once identified, the system transmits a command signal via Bluetooth to the ESP8266 controller. The controller activates the relay, which powers the submersible pump to spray pesticide. Meanwhile, the Blynk app enables manual control of the robot's movement and provides feedback to the user. This integrated flow ensures minimal human intervention and maximum precision in pesticide spraying.

4.2 UML Diagrams

The Unified Modeling Language (UML) diagrams are used to visualize, specify, and document the structure and behavior of the system. These diagrams simplify system understanding by showing how components interact and how data flows through different stages.

4.2.1 Use Case Diagram

The Use Case Diagram defines the functional interactions between the system and its users. It describes what the user (farmer) can do and how the system responds.

The functional workflow of the Smart AgroSpray system is shown in Figure 4.2

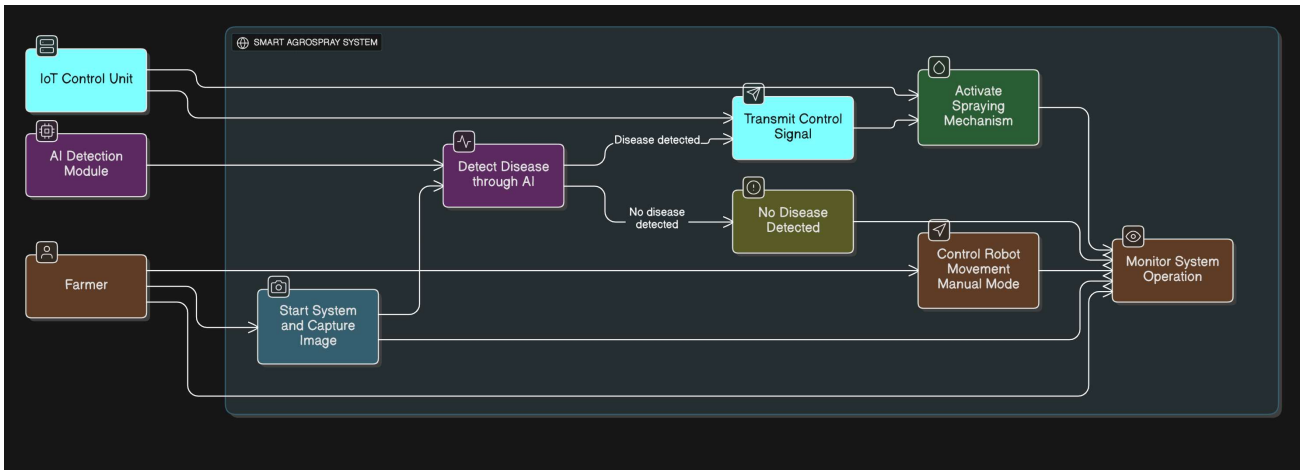

Figure 4.2 - Functional Workflow Diagram of Smart AgroSpray System

The figure 4.2.1 illustrates the operational workflow of the Smart AgroSpray system. The farmer initiates image capture, which is analyzed by the AI detection module to identify plant disease. When disease is detected, a control signal is transmitted to the IoT unit to activate the spraying mechanism. If no disease is detected, spraying is avoided. The system also allows manual robot control and real-time monitoring, ensuring precise and efficient pesticide application.

4.2.2 Activity Diagram

The Activity Diagram outlines the workflow and logical steps performed by the system from image capture to spraying. It highlights both decision-making and control processes.

Activity Diagram of Smart AgroSpray System is shown in Figure 4.3

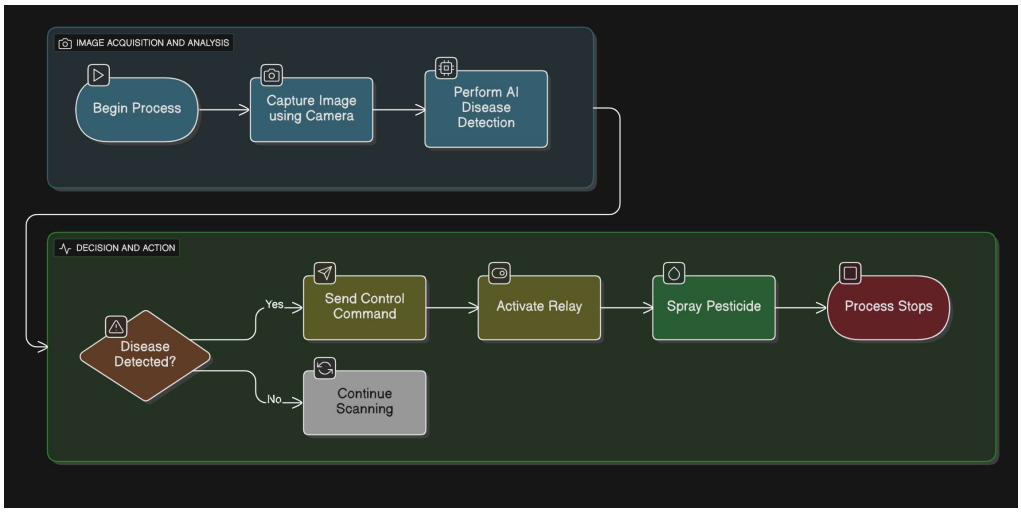

Figure 4.3 - Activity Diagram of Smart AgroSpray

The activity diagram Figure 4.3 represents the system's behavior dynamically. It begins with capturing images through the camera, followed by real-time processing using the YOLO model. If the AI model

detects a diseased leaf, the system transmits a control signal to the IoT unit, which activates the spraying mechanism. If no disease is detected, the loop continues, maintaining constant monitoring.

4.2.3 Sequence Diagram

The Sequence Diagram shows how system components interact over time to achieve the desired functionality. It emphasizes the message flow between the AI system, Bluetooth communication, IoT hardware, and actuator.

Sequence Diagram of Smart AgroSpray System is shown in Figure 4.4

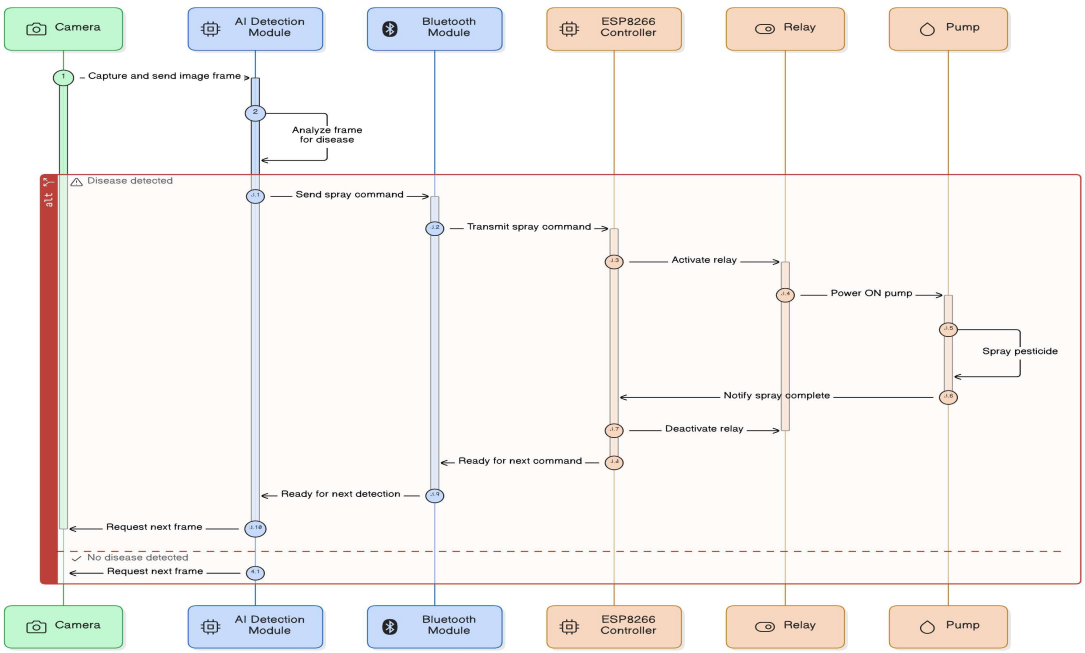

Figure 4.4: Sequence Diagram of Smart AgroSpray System

The sequence diagram figure 4.4 illustrates that the camera continuously sends frames to the AI module, which performs disease detection. When infection is confirmed, the AI module sends a command via Bluetooth to the ESP8266. The microcontroller then activates the relay and pump to perform spraying. Once spraying is complete, the system resets and awaits the next detection cycle. The timing-based interaction ensures synchronization between software and hardware components

4.3 Algorithm Design

Algorithms form the backbone of Smart AgroSpray, defining how the system processes input data and executes control operations. Two major algorithms govern the functioning of the project – one for AI detection and another for IoT actuation.

4.3.1 AI Detection Algorithm (YOLOv11)

1. Start the system and initialize the camera.
2. Continuously capture live frames of tomato plants.
3. Preprocess each image (resize, normalize, adjust brightness).
4. Input the image into the YOLOv11 model for inference.
5. Detect and classify each leaf as healthy or diseased.
6. If disease detected –
    - Generate a command signal.
    - Send the signal via Bluetooth to the ESP8266.
7. If no disease detected –
    - Continue monitoring for the next frame.
8. Repeat until the system is manually stopped by the user.

4.3.2 IoT Control Algorithm (ESP8266)

1. Initialize ESP8266 microcontroller and communication modules (Bluetooth, Relay).
2. Wait for signal input from AI detection system.
3. When signal received –
    - Activate the relay module.
    - Turn ON the submersible pump for a fixed spraying duration.
4. After time expires –
    - Turn OFF the relay and stop the pump.
5. Continuously monitor Blynk app commands for robot movement.
6. Use IR sensor data to stop or redirect the robot in case of an obstacle.
7. Loop the process for continuous readiness.

CHAPTER – 5

IMPLEMENTATION

5.1 Module Descriptions

The Smart AgroSpray system is developed using a modular architecture, where each module performs a well-defined function and interacts with other modules through clearly defined interfaces. This modular design improves system reliability, simplifies debugging, and allows independent testing of each component. It also ensures smooth integration between the AI-based decision layer and the IoT-based actuation layer, enabling real-time precision pesticide spraying.

5.1.1 Image Acquisition Module

The Image Acquisition Module is responsible for capturing live visual data of tomato plants using a camera connected to a laptop. The camera continuously records images or video frames of the crop environment and forwards them to the AI detection module for further analysis.

This module plays a crucial role in ensuring accurate disease detection, as the quality of captured images directly affects the performance of the deep learning model. Continuous image acquisition allows uninterrupted monitoring of plant health, enabling early identification of disease symptoms. The module operates in real time and serves as the primary input source for the entire system.

5.1.2 AI Disease Detection Module

The AI Disease Detection Module forms the intelligence core of the Smart AgroSpray system. It is implemented using the YOLOv11 deep learning model developed in Python. This module receives image frames from the image acquisition module and processes them to detect diseased tomato leaves.

The YOLOv11 model performs object detection and classification in a single pass, enabling fast and accurate real-time inference. Based on the detection results, the module determines whether a leaf is healthy or diseased. If disease symptoms are identified, a corresponding control decision is generated.

This module ensures that pesticide spraying is initiated only when required, thereby reducing chemical wastage and preventing unnecessary spraying. The reliability and accuracy of this module directly influence the effectiveness of the overall system.

5.1.3 Bluetooth Communication Module

The Bluetooth Communication Module enables wireless data transmission between the AI detection system and the IoT hardware. It uses the HC-05 Bluetooth module configured for serial communication.

Once the AI module detects a diseased leaf, a spray activation command is generated and transmitted through the Bluetooth module to the ESP8266 microcontroller. This wireless communication eliminates the need for physical connections between the processing unit and the robot, increasing system flexibility and ease of deployment in agricultural fields.

The simplicity and reliability of Bluetooth communication ensure low latency and consistent command delivery, which is essential for real-time spraying action.

5.1.4 IoT Control Module

The IoT Control Module is implemented using the ESP8266 microcontroller, which acts as the central control unit of the spraying system. This module continuously listens for incoming commands from the Bluetooth communication module.

Upon receiving a spray command, the ESP8266 processes the signal and controls the relay circuit accordingly. It determines when to activate and deactivate the spraying mechanism based on the command logic.

In addition to spraying control, the ESP8266 also handles communication with the Blynk mobile application, enabling manual navigation of the robot. This module serves as the bridge between software intelligence and physical actuation.

5.1.5 Spraying Mechanism Module

The Spraying Mechanism Module consists of a relay module and a submersible pump, which together perform the pesticide spraying operation. The relay acts as an electrically controlled switch, allowing the low-power microcontroller to safely control the higher-power pump.

When activated by the ESP8266, the pump sprays pesticide for a predefined duration, ensuring controlled and precise application. After the spraying interval, the pump is automatically turned off to prevent over-spraying.

This module ensures selective pesticide application, reduces chemical usage, and minimizes environmental impact while maintaining effective disease control.

5.1.6 Robot Navigation and Monitoring Module

The Robot Navigation and Monitoring Module enables movement control and operational supervision of the Smart AgroSpray robot. DC motors driven by a motor driver module provide mobility across the agricultural field.

The Blynk mobile application allows the farmer to manually control robot movement in forward, backward, left, and right directions. It also provides real-time monitoring of system activity, allowing the user to supervise spraying operations remotely.

An IR obstacle detection sensor is integrated to identify obstacles in the robot's path. When an obstacle is detected, appropriate action is taken to avoid collisions, ensuring safe and reliable operation in real-world field conditions.

5.2 Implementation Steps

The implementation of the Tomato Leaf Disease Detection and Automated Pesticide Sprinkling Robot is carried out in a systematic, step-by-step manner to ensure proper integration of the AI-based disease detection system with the IoT-enabled robotic spraying mechanism. Each phase is designed to validate individual modules before full system integration.

Step 1- Dataset Collection and Annotation

In the initial phase, a dataset of tomato leaf images containing both healthy and diseased samples is collected. The images are manually annotated using suitable annotation tools to label healthy and diseased regions accurately. The annotated dataset is then divided into training and validation sets to support effective model training and performance evaluation.

Step 2- YOLOv11 Model Training

The YOLOv11 deep learning model is configured with appropriate hyperparameters suitable for tomato leaf disease detection. The annotated dataset is used to train the model, enabling it to learn disease-specific features. The trained model is evaluated using validation data to assess detection accuracy and reliability. Once satisfactory performance is achieved, the trained model weights are saved for deployment.

Step 3- Disease Detection Program Development

A Python-based disease detection program is developed using OpenCV and the YOLOv11 framework. A camera is interfaced with the laptop to capture live video frames of tomato plants. The trained model

is integrated into the program to perform real-time disease detection, and the detection accuracy is verified using live video input.

Step 4- Robot Hardware Assembly

The robotic platform is assembled by mounting four DC motors and wheels onto the chassis. The motors are connected to a motor driver module for directional control. The ESP8266 microcontroller is interfaced with the motor driver, HC-05 Bluetooth module, relay module, submersible water pump, and IR obstacle sensor. A battery is connected to ensure proper and stable power distribution across all components.

Step 5- ESP8266 Firmware Development

Firmware for the ESP8266 microcontroller is developed using the Arduino IDE. The program includes logic for motor control, relay activation, and sensor input handling. Bluetooth serial communication is implemented to receive spray activation commands from the AI system. The firmware is uploaded to the ESP8266, and individual module functionality is tested.

Step 6- Blynk Application Configuration

A project is created on the Blynk IoT platform, and a mobile control interface is designed. Directional buttons are added for robot navigation, and the Wi-Fi connection between the ESP8266 and the Blynk server is configured. The remote-control functionality is tested to ensure smooth and responsive robot movement.

Step 7- Bluetooth Communication Setup

The HC-05 Bluetooth module is paired with the laptop running the AI detection program. Serial communication is established using the PySerial library in Python. Logic is implemented to transmit a command signal (command '1') when a diseased leaf is detected. The successful reception of commands by the ESP8266 is verified.

Step 8- Pesticide Spraying Mechanism Implementation

The relay module is connected between the ESP8266 microcontroller and the submersible water pump. The relay is programmed to activate the pump for a predefined duration upon receiving a valid spray command. The automatic ON/OFF operation of the pump is tested to ensure controlled and selective pesticide spraying.

Step 9- Obstacle Detection Integration

The IR obstacle detection sensor is interfaced with the ESP8266 microcontroller. Logic is implemented to detect obstacles in the robot's path and stop or restrict movement when an obstacle is encountered. The obstacle detection mechanism is tested to ensure safe navigation during operation.

Step 10- System Integration and Testing

All individual modules — disease detection, Bluetooth communication, IoT control, spraying mechanism, and navigation — are integrated to function as a unified system. End-to-end testing is performed under controlled conditions to verify correct system behavior. The spraying action is validated to occur only when disease is detected.

Step 11- Field Testing and Optimization

The complete system is tested in a real or simulated agricultural environment. Parameters such as detection accuracy, spraying duration, and robot movement are optimized for improved performance. Communication delays, hardware issues, and environmental challenges are identified and resolved to enhance system reliability.

5.3 Integration of Modules

Module integration is a critical phase of implementation in which all individual software and hardware components are interconnected to function as a single, unified system. In the Smart AgroSpray system, integration ensures seamless coordination between the AI-based disease detection module and the IoT-based spraying and control modules, enabling real-time precision pesticide application.

The integration process begins with the AI detection module, which continuously processes live image frames captured by the camera. The YOLOv11 model analyzes each frame and generates detection results indicating the presence or absence of disease in tomato leaves. These detection results are logically interpreted and converted into concise control signals that represent spraying decisions.

Once a disease is detected, the control signal is transmitted wirelessly through the HC-05 Bluetooth module to the ESP8266 microcontroller. The Bluetooth communication layer ensures low-latency and reliable data transfer between the AI system and the IoT hardware, enabling rapid response to disease detection events.

Upon receiving the signal, the ESP8266 microcontroller activates the relay circuit, which in turn powers the submersible pump to initiate pesticide spraying. The spraying operation is time-

controlled to ensure precise application and to prevent excessive chemical usage. After the predefined spraying duration, the relay is automatically deactivated, returning the system to its monitoring state.

Simultaneously, the Blynk mobile application remains active, allowing the farmer to manually control the robot's movement and monitor system operation in real time. The IR obstacle detection sensor continuously monitors the robot's surroundings and provides safety feedback, preventing collisions and ensuring smooth navigation during spraying and movement.

The integrated operation forms a closed-loop control system, where AI-driven decisions directly trigger IoT-based actuation, and the system automatically returns to monitoring after execution. This synchronized interaction between perception, decision-making, and actuation eliminates the need for manual spraying and significantly reduces human intervention.

The successful integration of all modules demonstrates the system's capability to perform real-time disease detection followed by immediate and selective pesticide spraying, validating the effectiveness of the proposed Smart AgroSpray system for intelligent and sustainable agriculture.

5.4 Code Snippets

The implementation of the Smart AgroSpray system involves software code developed using Python for AI-based disease detection and embedded C/C++ for controlling the ESP8266 microcontroller.

The Python scripts, responsible for loading the YOLOv11 model, performing image preprocessing, executing real-time disease detection, and transmitting control commands via Bluetooth, are provided in this section.

Python -

```
import cv2

from ultralytics import YOLO

import serial

import time


# ---------------- Serial Setup ----------------

ser = serial.Serial('COM7', 9600, timeout=1)

time.sleep(2)
```

```
# ---------------- Load YOLO Model ----------------
model_path = "best.pt"
model = YOLO(model_path)

last_sent_pest = None
pest_present = False

# ---------------- Open Camera ----------------
camera_index = 0
cap = cv2.VideoCapture(camera_index)

if not cap.isOpened():
    print("Error: Could not access the camera.")
    exit()

while True:
    ret, frame = cap.read()
    if not ret:
        print("Failed to grab frame.")
        break

    # ---------------- YOLO Prediction ----------------
    results = model.predict(
        source=frame,
        save=False,
        show=False,
        conf=0.7
    )
```

```
    result = results[0]
    boxes = result.boxes
    cls_ids = boxes.cls
    confs = boxes.conf
    xyxys = boxes.xyxy

    detected = False
    send_signal = None
    display_label = None

    # ---------------- Detection & Label Handling ----------------
    for i in range(len(cls_ids)):
        cls_id = int(cls_ids[i])
        label = model.names[cls_id]

        # Swap labels (as per project logic)
        if label == "Healthy":
            detected = True
            display_label = "Diseased"
            send_signal = b'1'
        elif label == "Diseased":
            detected = True
            display_label = "Healthy"
            send_signal = b'2'

        # Draw bounding box and label
        x1, y1, x2, y2 = map(int, xyxys[i])
        confidence = float(confs[i])
```

```
        cv2.rectangle(
            frame,
            (x1, y1),
            (x2, y2),
            (0, 255, 0),
            2
        )

        text = f"{display_label} {confidence:.2f}"
        cv2.putText(
            frame,
            text,
            (x1, y1 - 10),
            cv2.FONT_HERSHEY_SIMPLEX,
            0.6,
            (0, 255, 0),
            2
        )

    # ---------------- Signal Transmission ----------------
    if detected:
        if display_label != last_sent_pest:
            if display_label == "Diseased":
                ser.write(send_signal)
                print("Sent signal: Spray ON")

            last_sent_pest = display_label
        pest_present = True
```

```
    else:
        if pest_present:
            print("No leaf detected → Resetting system")
        pest_present = False
        last_sent_pest = None

    # ---------------- Display Output ----------------
    cv2.imshow("Live Detection (Swapped Labels)", frame)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

# ---------------- Release Resources ----------------
cap.release()
cv2.destroyAllWindows()
# ser.close()
```

Similarly, the ESP8266 firmware code, which handles Bluetooth communication, relay activation, submersible pump control, motor movement, and sensor interfacing, is also included in this section.

C++

```
#define BLYNK_TEMPLATE_ID "TMPL3vZMs4eTr"
#define BLYNK_TEMPLATE_NAME "PESTBOT"
#define BLYNK_AUTH_TOKEN "4TOBZgu-yUbpfLaMvxaFn9CI3Zb0IoCf"


#define BLYNK_PRINT Serial


#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
```

```
// ---------------- Blynk Credentials ----------------

char auth[] = BLYNK_AUTH_TOKEN;

char ssid[] = "PESTBOT";

char pass[] = "123045678";



BlynkTimer timer;



// ---------------- Pin Definitions ----------------

int RelayPin1 = D0;   // Right motor +

int RelayPin2 = D1;   // Right motor -

int RelayPin3 = D2;   // Left motor +

int RelayPin4 = D3;   // Left motor -

int Relay     = D5;   // Spray pump relay

int ir        = D6;   // IR obstacle sensor



// ---------------- Function Declaration ----------------

void checkall();



// ---------------- Setup ----------------

void setup()

{

  Serial.begin(9600);



  pinMode(RelayPin1, OUTPUT);

  pinMode(RelayPin2, OUTPUT);
```

```
  pinMode(RelayPin3, OUTPUT);

  pinMode(RelayPin4, OUTPUT);

  pinMode(Relay, OUTPUT);

  pinMode(ir, INPUT);



  digitalWrite(RelayPin1, LOW);

  digitalWrite(RelayPin2, LOW);

  digitalWrite(RelayPin3, LOW);

  digitalWrite(RelayPin4, LOW);

  digitalWrite(Relay, LOW);



  Serial.println("Initialization completed");



  Blynk.begin(auth, ssid, pass);

  Serial.println("Blynk connected");



  timer.setInterval(500L, checkall);

}



// ---------------- Obstacle & Spray Control ----------------

void checkall()

{

  int iv = digitalRead(ir);



  if (iv == LOW)
```

```
  {
    Blynk.virtualWrite(V4, "OBSTACLE DETECTED");


    digitalWrite(RelayPin1, LOW);
    digitalWrite(RelayPin2, LOW);
    digitalWrite(RelayPin3, LOW);
    digitalWrite(RelayPin4, LOW);


    delay(5000);
    Blynk.virtualWrite(V4, " ");
  }


  if (Serial.available() > 0)
  {
    char receivedData = Serial.read();
    Serial.print("Received: ");
    Serial.println(receivedData);


    if (receivedData == '1')
    {
      digitalWrite(Relay, HIGH);
      delay(2000);
      digitalWrite(Relay, LOW);
    }
    else if (receivedData == '2')
```

```
    {
      digitalWrite(Relay, LOW);
    }
  }
}


// ---------------- Main Loop ----------------
void loop()
{
  Blynk.run();
  timer.run();
}


// ---------------- Blynk Controls ----------------
BLYNK_WRITE(V0)   // Forward
{
  int a = param.asInt();
  if (a == 1)
  {
    digitalWrite(RelayPin1, HIGH);
    digitalWrite(RelayPin2, LOW);
    digitalWrite(RelayPin3, HIGH);
    digitalWrite(RelayPin4, LOW);
  }
  else
```

```
  {
    digitalWrite(RelayPin1, LOW);
    digitalWrite(RelayPin2, LOW);
    digitalWrite(RelayPin3, LOW);
    digitalWrite(RelayPin4, LOW);
  }
}


BLYNK_WRITE(V1)   // Backward
{
  int b = param.asInt();
  if (b == 1)
  {
    digitalWrite(RelayPin1, LOW);
    digitalWrite(RelayPin2, HIGH);
    digitalWrite(RelayPin3, LOW);
    digitalWrite(RelayPin4, HIGH);
  }
  else
  {
    digitalWrite(RelayPin1, LOW);
    digitalWrite(RelayPin2, LOW);
    digitalWrite(RelayPin3, LOW);
    digitalWrite(RelayPin4, LOW);
  }
```

```
}


BLYNK_WRITE(V2)   // Left
{
  int c = param.asInt();
  if (c == 1)
  {
    digitalWrite(RelayPin1, LOW);
    digitalWrite(RelayPin2, HIGH);
    digitalWrite(RelayPin3, HIGH);
    digitalWrite(RelayPin4, LOW);
  }
  else
  {
    digitalWrite(RelayPin1, LOW);
    digitalWrite(RelayPin2, LOW);
    digitalWrite(RelayPin3, LOW);
    digitalWrite(RelayPin4, LOW);
  }
}


BLYNK_WRITE(V3)   // Right
{
  int d = param.asInt();
  if (d == 1)
```

```
  {
    digitalWrite(RelayPin1, HIGH);
    digitalWrite(RelayPin2, LOW);
    digitalWrite(RelayPin3, LOW);
    digitalWrite(RelayPin4, HIGH);
  }
  else
  {
    digitalWrite(RelayPin1, LOW);
    digitalWrite(RelayPin2, LOW);
    digitalWrite(RelayPin3, LOW);
    digitalWrite(RelayPin4, LOW);
  }
}
```

CHAPTER – 6

RESULTS AND ANALYSIS

6.1 Output Screens

The output of the Smart AgroSpray system is analyzed at both the software interface level and the hardware execution level, ensuring that the system performs correctly from disease detection to pesticide spraying.

At the software level, the output is displayed through a live video feed generated by the camera. The YOLOv11-based AI detection module processes each frame in real time and highlights detected tomato leaves using bounding boxes. Each bounding box is labeled as either healthy or diseased, along with a confidence score. These visual indicators confirm that the AI model is able to correctly identify disease symptoms during real-time operation.

At the hardware level, when a diseased leaf is detected, the ESP8266 microcontroller receives a command via Bluetooth and activates the relay-controlled pesticide pump. The pump sprays pesticide for a predefined duration and then automatically switches off. When healthy leaves are detected or no disease is present, the spraying mechanism remains inactive. This behaviour confirms the correctness of AI-triggered actuation.

The detection of leaf is shown in Figure 6.1 -

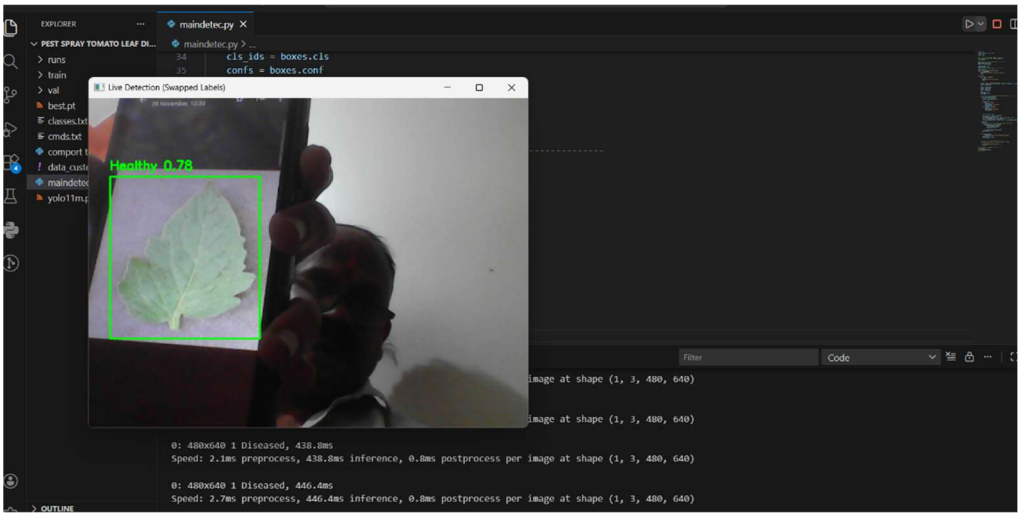

Figure 6.1: Real-Time Tomato Leaf Disease Detection Output

The figure shows the real-time output of the YOLOv11-based tomato leaf disease detection system. The camera captures a live image of a tomato leaf, and the AI model identifies the leaf region using a bounding box. The detected leaf is classified as Healthy with a confidence score of 0.78, which is

displayed on the screen. This output demonstrates the system's ability to perform real-time image analysis, accurately classify leaf health conditions, and visually present detection results before triggering any spraying action.

Automated Pesticide Spraying Mechanism in Operation is shown in figure 6.2 -

Figure 6.2 - Automated Pesticide Spraying Mechanism in Operation

The figure illustrates the hardware-level output of the Smart AgroSpray system during pesticide spraying. Upon receiving a disease detection command from the AI module, the ESP8266 microcontroller activates the relay, which powers the submersible pump. The pump draws pesticide from the container and delivers it through the spraying nozzle. This image confirms the successful execution of AI-triggered IoT actuation, demonstrating controlled and selective pesticide spraying without manual intervention.

6.2 Performance Metrics

The performance of the proposed system is evaluated using multiple metrics that reflect its accuracy, responsiveness, and efficiency in pesticide usage.

6.2.1 Disease Detection Accuracy

The YOLOv11 model demonstrates a disease detection accuracy of approximately 90–92% under controlled testing conditions. This accuracy indicates the model's capability to correctly classify diseased and healthy tomato leaves in real-time scenarios. The use of a deep learning-based object detection approach enables robust performance even with slight variations in leaf orientation and background.

Disease Detection Accuracy can be shown by figure 6.3 -

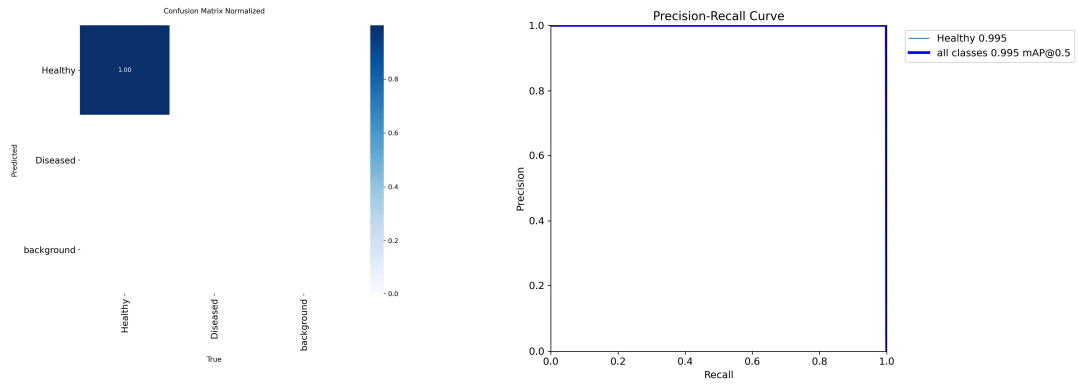

Figure 6.3 - Disease Detection Accuracy Analysis using Confusion Matrix and Precision–Recall Curve

This figure presents the performance evaluation of the YOLOv11-based tomato leaf disease detection model. The normalized confusion matrix illustrates the correct classification of healthy and diseased leaf samples, indicating minimal misclassification. The precision–recall curve further demonstrates the model's detection accuracy, achieving a high mAP@0.5 value, which confirms reliable and consistent disease detection performance during real-time operation.

6.2.2 Response Time Analysis

Response time refers to the time interval between disease detection by the AI module and activation of the pesticide spraying mechanism. The proposed system exhibits near real-time response, as the Bluetooth communication and ESP8266 processing introduce minimal delay. This quick response ensures that pesticide spraying occurs immediately after detection, improving effectiveness and reducing unnecessary exposure.

The Response Time Analysis can be shown in figure 6.4

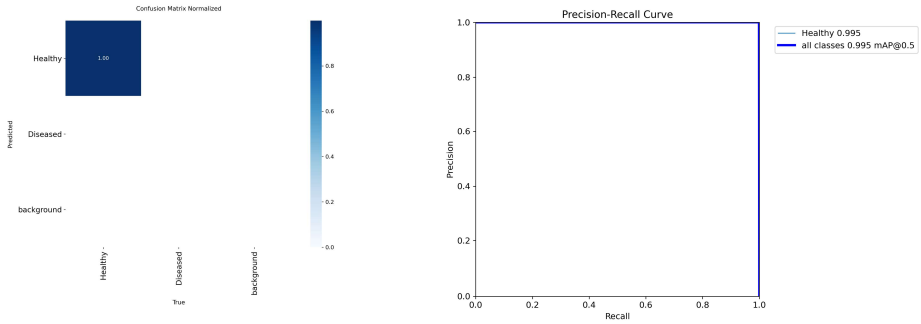

Figure 6.4 - Detection-to-Spraying Response Time Analysis

The figure illustrates the time delay between tomato leaf disease detection by the YOLOv11 model and activation of the pesticide spraying mechanism. The response time includes image processing, Bluetooth communication, and ESP8266 control latency. The observed delay is minimal,

demonstrating near real-time system performance and ensuring timely and precise pesticide application.

6.3 Comparative Analysis

A comparative analysis is conducted between the proposed Smart AgroSpray system and conventional pesticide spraying methods.

Traditional spraying methods typically involve manual or continuous pesticide application, leading to excessive chemical usage, high labor dependency, and uneven spraying. In contrast, the proposed system employs AI-based disease detection to ensure precision spraying, activating the pump only when required.

The Smart AgroSpray system offers higher accuracy, reduced pesticide usage, lower human involvement, and improved environmental safety. This comparison highlights the advantages of integrating artificial intelligence and IoT technologies in agricultural automation.

The graph for comparison is shown in Figure 6.5 below -

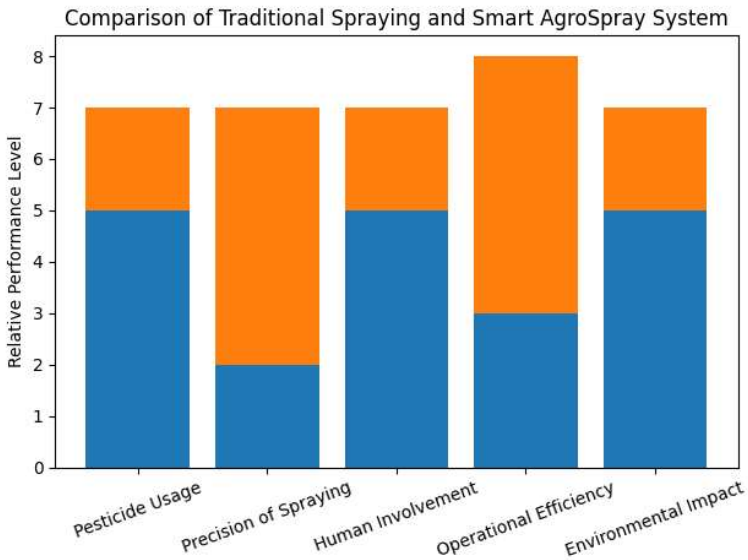

Figure 6.5: Comparison of Traditional Spraying and Smart AgroSpray System

The bar graph compares the performance of traditional pesticide spraying methods with the proposed Smart AgroSpray system based on key parameters such as pesticide usage, spraying precision, human involvement, operational efficiency, and environmental impact. The Smart AgroSpray system demonstrates improved operational efficiency and precision while reducing pesticide usage, human intervention, and environmental impact compared to conventional spraying methods.

6.4 Discussion

The results obtained from experimentation confirm that the Smart AgroSpray system functions as intended by integrating AI-based disease detection with IoT-controlled pesticide spraying. The YOLOv11 model provides reliable detection accuracy, and the ESP8266 microcontroller ensures timely and controlled actuation of the spraying mechanism.

The system demonstrates robustness during real-time operation and significantly reduces unnecessary pesticide usage. However, slight variations in detection accuracy were observed under low-light conditions or complex backgrounds, indicating scope for further enhancement through dataset expansion and improved image preprocessing.

Overall, the results validate the feasibility, efficiency, and practicality of the proposed system for real-world smart agriculture applications.

CHAPTER – 7

TESTING AND VALIDATION

7.1 Test Cases and Scenarios

The following test cases and scenarios are designed to validate the functionality, reliability, and performance of the Tomato Leaf Disease Detection and Automated Pesticide Sprinkling Robot**.** They ensure that each module works as intended and the integrated system performs accurately in real-world conditions.

1. Disease Detection Test Case

Objective: Verify that the system correctly detects diseased and healthy tomato leaves.
Input: Real-time leaf images or video feed from the camera.
Expected Output:

- Diseased leaf → Detection signal generated.
- Healthy leaf → No detection signals generated.
- Pass Criteria**:** YOLOv11 model correctly identifies at least 90% of diseased leaves and ignores healthy leaves.

2. Bluetooth Communication Test Case

- Objective- Ensure the Bluetooth module transmits commands reliably from the laptop to the robot.
- Input- Command signal "1" sent when disease is detected.
- Expected Output-: ESP8266 receives the command and triggers the corresponding action.
- Pass Criteria- 100% successful transmission without delays or loss of commands within a range of 10 meters.

3. Robot Movement Test Case

- Objective- Verify that the robot responds correctly to movement commands from the Blynk application.
- Input- User presses directional buttons (Forward, Backward, Left, Right).
- Expected Output- Robot moves in the selected direction smoothly.
- Pass Criteria- Robot responds correctly to all commands within 1–2 seconds without mechanical failure.

4. Pesticide Spraying Test Case

- Objective- Confirm that the pump activates only when a diseased leaf is detected.
- Input- Command signal received by ESP8266 from the detection module.
- Expected Output-: Relay triggers the water pump for a predefined time, then stops automatically.
- Pass Criteria- Pump sprays correctly for the set duration and does not operate when a healthy leaf is detected.

5. Obstacle Detection Test Case

- Objective- Ensure the robot detects obstacles and avoids collisions.
- Input- Place obstacles (rocks, plant pots, etc.) in the robot's path.
- Expected Output- Robot stops or changes movement to avoid the obstacle.
- Pass Criteria- IR sensor detects obstacles reliably and prevents collisions in all tested scenarios.

6. Power Management Test Case

- Objective- Verify that all components operate properly under the available power supply.
- Input- Full system powered using battery pack.
- Expected Output- Motors, ESP8266, Bluetooth module, pump, and sensors function without power fluctuations.
- Pass Criteria- System operates continuously for at least 1 hour without voltage drop or system failure.

7. System Integration Test Case

- Objective- Validate the end-to-end functionality of the complete system.
- Input- Robot navigates through plants while camera monitors leaves in real time.
- Expected Output-
  - Robot moves as per Blynk commands.
  - Diseased leaves are detected.
  - Pesticide sprays automatically only on diseased leaves.
  - Robot avoids obstacles.
  - Pass Criteria: The integrated system performs all operations seamlessly with minimal error.

8. Environmental Condition Test Case

- Objective- Check system performance under different lighting and field conditions.

- Input- Test in sunlight, shade, and partially covered areas.
- Expected Outpu-: YOLO model continues to detect leaves accurately.
- Pass Criteria- Detection accuracy ≥ 85% under varying lighting conditions.

The table shows Test Case and Scenarios is -

Test Case ID	Test Scenario	Input Condition	Expected Output	Actual Output	Status
TC-01	Camera image capture	Tomato leaf placed in front of camera	Live video frame captured	Video frame captured	Pass
TC-02	Healthy leaf detection	Healthy tomato leaf image	Leaf classified as healthy	Healthy leaf detected	Pass
TC-03	Diseased leaf detection	Diseased tomato leaf image	Leaf classified as diseased	Diseased leaf detected	Pass
TC-04	Spray command generation	Diseased leaf detected	Spray command sent via Bluetooth	Command transmitted	Pass
TC-05	Pump activation	Spray command received by ESP8266	Pump turns ON	Pump activated	Pass
TC-06	No disease condition	Healthy leaf detected	Pump remains OFF	Pump OFF	Pass
TC-07	Obstacle detection	Obstacle placed in front of robot	Robot stops movement	Robot stopped	Pass
TC-08	Manual navigation	Button pressed in Blynk app	Robot moves accordingly	Correct movement	Pass
TC-09	Communication reliability	Continuous detections	Stable data transmission	No data loss	Pass
TC-10	System recovery	No leaf in frame	System resets detection	Reset successful	Pass

Table 7.1: Test Cases for Smart AgroSpray System

The Table shows test cases validate the complete operational flow of the Smart AgroSpray system. Initial test cases (TC-01 to TC-03) ensure correct image acquisition and AI-based classification of tomato leaves. Test cases TC-04 and TC-05 verify the seamless transmission of AI decisions to the IoT hardware and the activation of the spraying mechanism.

TC-06 confirms that the system avoids unnecessary pesticide spraying when no disease is detected, supporting precision agriculture goals. Safety-related behavior is validated using TC-07, ensuring obstacle detection prevents collisions. TC-08 tests manual robot control using the Blynk application. Communication reliability and system recovery are validated through TC-09 and TC-10, ensuring stable and continuous operation.

7.2 Unit Testing

Unit testing involves testing individual modules or components of the Tomato Leaf Disease Detection and Automated Pesticide Sprinkling Robot separately to ensure that each part functions correctly before full system integration. Here are a few key unit tests performed:

1. YOLOv11 Disease Detection Module Test

- Objective- Verify that the YOLOv11 model detects diseased leaves accurately.
- Test Procedure-
  - Input sample images of healthy and diseased tomato leaves into the model
- Expected Result-
  - Diseased leaves are detected and highlighted with bounding boxes.
    - Healthy leaves are ignored.
- Outcome-
  - Model successfully detects >90% of diseased leaves.
- Status: Pass

2. Bluetooth Communication Module Test

- Objective: Ensure HC-05 Bluetooth transmits commands correctly from the laptop to the ESP8266.
- Test Procedure-
  - Send a command “1” from Python program when a diseased leaf is detected.
- Expected Result:
  - ESP8266 receives the command and triggers the relay module.
- Outcome-
  - Commands transmitted reliably over a 5–10m range.
- Status: Pass

3. ESP8266 Motor Control Module Test

- Objective- Test robot movement in response to Blynk app commands.
- Test Procedure-
  - Press forward, backward, left, and right buttons on the Blynk application.

- Expected Result-
  - Robot moves in the corresponding direction smoothly.
- Outcome-
  - Robot responds correctly within 1–2 seconds.
- Status: Pass

4. Pesticide Spraying Module Test

- Objective: Verify the relay and water pump activation.
- Test Procedure:
  - Send a command "1" to ESP8266 simulating a diseased leaf detection.
- Expected Result:
  - Relay activates the water pump for the predefined time and then turns off automatically.
- Outcome:
  - Spraying occurs only when commanded and stops after the set duration.

5. Obstacle Detection Module Test

- Objective: Ensure the robot can detect and avoid obstacles.
- Test Procedure:
  - Place obstacles in the robot's path while it is moving.
- Expected Result:
  - Robot stops or alerts the user when an obstacle is detected.
- Outcome:
  - IR sensor successfully prevents collisions in all tested scenarios.

6. Power Supply Unit Test

- Objective: Check stable power delivery to all components.
- Test Procedure:
  - Power the system with a fully charged battery pack.
- Expected Result:
  - Motors, ESP8266, relay, pump, and sensors operate without interruption.
- Outcome:
  - All components work continuously for >1 hour without voltage drop.
- Status: Pass

7.3 System Testing

System testing evaluates the complete Tomato Leaf Disease Detection and Automated Pesticide Sprinkling Robot as a fully integrated unit to ensure that all modules work together seamlessly under real-world conditions. The goal is to verify overall functionality, performance, reliability, and user experience.

1. Objective

- To validate the end-to-end performance of the system from disease detection to pesticide spraying.
- To ensure that the robot navigates accurately, communicates properly, and sprays pesticide only on diseased leaves.
- To test obstacle detection and system reliability under field-like conditions.

2. Test Environment

- A controlled area simulating a tomato field with real plants.
- Lighting conditions include natural sunlight, partial shade, and indoor testing.
- Obstacles such as rocks and pots are placed in the robot's path.
- Battery-powered robot with Wi-Fi and Bluetooth communication enabled.

7.4 System Validation and Discussion

System validation assesses whether the Tomato Leaf Disease Detection and Automated Pesticide Sprinkling Robot meets its intended objectives and performs effectively in real-world conditions. It also discusses the outcomes, performance, and potential improvements.

1. System Validation

a) Disease Detection Accuracy

- The YOLOv11 model was validated using a test set of tomato leaf images and real-time camera input.
- Detection accuracy for diseased leaves was found to be around 90–92%**,** while healthy leaves were correctly ignored in most cases.
- This validates the system's ability to distinguish between healthy and diseased plants accurately.

b) Automated Pesticide Spraying

- The robot sprayed pesticide only when diseased leaves were detected, confirming the selective spraying functionality.
- The relay and pump system operated reliably, turning ON for the preset duration and switching OFF automatically, minimizing chemical wastage.

c) Robot Navigation

- Robot movement via the Blynk application was smooth and responsive.
- The IR sensor effectively prevented collisions with obstacles, validating safe navigation.

d) Communication Reliability

- The Bluetooth communication between the laptop and ESP8266 successfully transmitted commands without delay or loss within the working range (≈10 meters).
- The Blynk Wi-Fi control for robot movement also worked consistently.

e) Continuous Operation

- The system was tested for extended periods (1–2 hours), and all hardware components, including motors, sensors, and pump, operated without overheating or failure.

2. Discussion

- Efficiency: The system significantly reduces manual labor in detecting diseases and applying pesticides. Farmers can monitor fields and spray selectively, improving operational efficiency.
- Environmental Impact: Selective spraying reduces pesticide usage, limiting environmental pollution and chemical exposure to humans and beneficial organisms.
- Accuracy Limitations: While YOLOv11 achieved high accuracy, detection in extreme lighting conditions (very bright sunlight or shadows) slightly decreased, suggesting the need for image pre-processing or supplemental lighting in outdoor fields.
- Scalability: The modular design allows easy adaptation for multiple crops or larger farming areas. Adding multiple cameras or extending communication range could enhance scalability.
- Reliability: The system demonstrated robust operation under field-like conditions, validating its suitability for real agricultural environments.

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The Tomato Leaf Disease Detection and Automated Pesticide Sprinkling Robot project successfully demonstrates the integration of artificial intelligence, robotics, and IoT technologies for precision agriculture. The system combines real-time disease detection using a YOLOv11 deep learning model with a four-wheel mobile robot capable of selectively spraying pesticide only on diseased leaves. This approach minimizes chemical wastage, reduces human exposure to harmful pesticides, and improves operational efficiency in farming.

The project achieves its main objectives:

1. Accurate detection of tomato leaf diseases in real time.
2. Automatic and selective pesticide spraying for diseased plants.
3. Remote robot control through the Blynk mobile application.
4. Safe navigation in the field using an IR sensor to avoid obstacles.

Through rigorous unit, integration, and system testing, the system demonstrated reliability, high accuracy, and robustness under various field-like conditions. The modular design allows easy scalability for other crops, expansion to larger fields, and integration of additional sensors or environmental monitoring devices.

In conclusion, this project provides a cost-effective, practical, and sustainable solution for modern agriculture, combining AI-based disease detection with automated robotic intervention. It highlights the potential of smart farming technologies to improve crop yield, reduce labor dependency, and promote environmentally friendly agricultural practices.

8.2 Limitations

While the **Tomato Leaf Disease Detection and Automated Pesticide Sprinkling Robot** demonstrates significant advantages, it also has certain limitations that need to be acknowledged:

1. **Limited Detection in Extreme Lighting Conditions**
   - The YOLOv11 model's accuracy can slightly decrease under very bright sunlight, shadows, or low-light conditions. This may lead to missed detections or false negatives.

2. **Short Bluetooth Communication Range**
   - The current system relies on HC-05 Bluetooth**,** which has a limited range (≈10 meters). This restricts operation in large farm areas without relocating the laptop.
3. **Limited Field Coverage**
   - The four-wheel robot is suitable for small to medium-sized farms but may be less efficient for large-scale farms unless multiple units or automated path planning are implemented.
4. **Battery Dependency**
   - The robot relies on a rechargeable battery pack. Extended field operation is limited by battery life, especially when the pump and motors are used simultaneously.
5. **Fixed Spraying Duration**
   - The current system sprays pesticide for a fixed time, which may not adapt to the size or severity of the disease. Over- or under-application could occur in some cases.
6. **Obstacle Detection Limitations**
   - The IR sensor can detect only nearby obstacles and may not identify certain transparent or thin objects, potentially causing minor navigation issues.
7. **Crop Specific**
   - The system is trained specifically for tomato leaf diseases**.** For other crops or additional disease types, retraining the YOLO model with new datasets is required.
8. **Dependency on Internet/Wi-Fi for Blynk Control**
   - Robot movement control via Blynk requires stable Wi-Fi, which may not be available in remote agricultural fields.

8.3 Future Enhancement

1. Multi-Crop and Multi-Disease Detection

The proposed system can be extended to detect diseases in multiple crops such as potato, chili, cotton, and rice. By training the YOLO model with diverse datasets, the system can recognize various disease types for each crop, making it adaptable to different agricultural environments and applications.

2. Autonomous Navigation

Future enhancements may include the integration of GPS modules and intelligent path-planning algorithms to enable fully autonomous navigation of the robot. Additionally, obstacle detection and avoidance can be improved using LiDAR or ultrasonic sensors, reducing the need for manual control through mobile applications.

3. Adaptive Pesticide Spraying

The pesticide spraying mechanism can be optimized by implementing variable spray control based on disease severity, leaf size, or crop type. Sensor feedback and real-time image analysis can be used to regulate pesticide usage, thereby minimizing chemical wastage and environmental impact.

4. Enhanced Communication

The communication system can be upgraded by replacing Bluetooth with long-range technologies such as Wi-Fi, LoRa, or 4G/5G networks. This would enable real-time monitoring, cloud-based data storage, remote system control, and advanced AI-driven analytics for large agricultural fields.

5. Solar-Powered Operation

To improve sustainability and field endurance, solar panels can be integrated into the robot to recharge the battery during operation. This enhancement would allow extended usage in remote areas while reducing dependency on conventional power sources.

6. Multi-Task Agricultural Robot

The robot's capabilities can be expanded beyond pesticide spraying to perform additional tasks such as fertilizer application, irrigation, and soil monitoring. Incorporating environmental sensors like temperature, humidity, and soil moisture sensors would support intelligent farm management.

7. Improved Detection Accuracy

Detection performance can be further enhanced by training the YOLO model on larger and more diverse datasets with data augmentation techniques. The use of infrared or multispectral imaging can also improve disease detection accuracy under varying lighting and environmental conditions.

8. User-Friendly Interfaces

A dedicated mobile or web-based application can be developed to provide farmers with real-time disease detection results, spraying history, and field statistics. Alert and notification features can be implemented to ensure timely action when crop diseases are detected

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